

PROJECT DOCUMENTATION

Automated Network Request Management in ServiceNow

A ServiceNow Service Catalog based solution for automated request intake, routing, approval, notification, tracking, and fulfilment

1. Project Overview

The Network Request Management system in ServiceNow automates request intake, routing, and fulfilment. Through dynamic forms, role-based approvals, and email notifications, it ensures transparency and efficiency. Optional automation integration further reduces manual workload and risk, enabling faster and more reliable network operations.

The project focuses on developing an automated Network Request Management system using ServiceNow. Its main purpose is to provide users with a simple and structured way to submit network-related service requests and to automate the complete process, from request submission through to approval, task creation, fulfilment, and notification. The system uses the Service Catalog, Catalog Item, Catalog Variables, Flow Designer, Approvals, a Network Task, Business Rules, Notifications, Access Controls, and Update Sets to manage the network request lifecycle. Instead of processing network requests manually, the configured workflow automatically moves the request through the required stages and keeps the concerned users informed about its progress.

1.1 Problem Statement

Manual and inconsistent handling of network requests can lead to delays, incomplete information, approval gaps, and poor visibility into request status. Without a structured intake and routing mechanism, requests are prone to missing data, inconsistent approval handling, and limited traceability for requesters, approvers, and fulfillers alike.

1.2 Objectives

- Provide a single, structured entry point for submitting network requests through the ServiceNow Service Catalog and Service Portal.
- Enforce mandatory field validation and auto-population to improve data quality at the point of intake.
- Route requests dynamically for approval based on request characteristics such as sensitivity and department.
- Automate task creation, status tracking, and email notifications across the request lifecycle.

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- Maintain data integrity through structured custom tables and approval state enforcement.
 - Keep the solution maintainable by relying primarily on Flow Designer rather than custom scripting.

1.3 Stakeholders

The network request lifecycle involves four principal stakeholders, each with a distinct set of responsibilities:

- **Requester / End User:** Submits the Network Request through the Service Portal and provides the required information.
- **Manager / Approver:** Reviews the request and provides the required approval before fulfilment.
- **Network Fulfilment Team:** Processes the generated Network Task and performs the required network-related activity.
- **ServiceNow Administrator:** Configures the catalog item, variables, workflows, Business Rules, Access Controls, Notifications, and other ServiceNow components.

1.4 Functional Scope

The functional scope of the project includes:

- Creation of the Network Request Service Catalog item.
- Configuration of request variables.
- Validation of mandatory information.
- Automatic population of the Requested For field.
- Creation of the required database records.
- Configuration of approval requests.
- Automated creation of Network Tasks.
- Sending email notifications.
- Updating request/task status.
- Applying Access Controls.
- Testing the complete request lifecycle.
- Maintaining data integrity.

1.5 System Architecture / Workflow

The end-to-end workflow of the Network Request Management system is as follows:

1. User opens the Network Request catalog item through the Service Portal.
2. User enters the required network request details such as Email ID, Phone Number, Device Type, Connection Type, Address, and Device Details.
3. The system validates the mandatory fields before submission.
4. The Requested For field is automatically populated with the current user.
5. User submits the request using Order Now.
6. ServiceNow creates the corresponding Request (REQ) and Requested Item (RITM) records.
7. The configured Flow Designer flow retrieves the catalog variables and creates the corresponding Network Database record.
8. The request is routed through the configured approval process.
9. Once the request is approved, the workflow creates a Network Task for fulfilment.
10. Notifications are generated at the required stages to inform the concerned users.
11. The Network Task is processed and the request status is updated.
12. The completed request can be tracked through ServiceNow.

The screenshot displays the 'Network Request' catalog item form in the ServiceNow Service Portal. The breadcrumb trail at the top reads: Home > Service Catalog > Network > Network Request. A search bar is located in the top right corner. The form is titled 'Network Request' with the subtitle 'Network Request Management'. A legend indicates that red asterisks (*) denote required fields. The 'Requester Information' section includes fields for 'Email ID' (required), 'Opened on behalf of' (a dropdown menu), and 'Phone Number' (required). There is an 'Upload' button for 'Proof of document' and a 'UserName' field. The 'type of device' is set to 'Laptop' in a dropdown menu. A field labeled 'please_provide_address_here' is present at the bottom. On the right side, there is a 'Quantity' dropdown set to '1', a 'Delivery Time' of '0 Days', and buttons for 'Add to Cart', 'Save as Draft', and 'Order Now'. Below these, a 'Required information' section lists 'Email ID', 'Phone Number', and 'Address' as required, with a 'Device Details' link. The top navigation bar includes links for Knowledge, Catalog, Requests, System Status, Cart, Tours, and a user profile for 'System Administrator'.

The Network Request catalog item as presented to the requester on the Service Portal.

1.6 Final Outcome

The completed system provides a structured and automated method for handling network service requests. The main outcome is a connected workflow in which request submission, data capture, validation, approval, Network Task creation, fulfilment, status update, and notification are all managed within ServiceNow. The project was tested using actual ServiceNow records, catalog forms, approval records, Flow Designer execution, and Network Task outputs. The screenshots included throughout this document are taken directly from the configured Personal Developer Instance (PDI) and serve as evidence of the implemented configuration and its results.

2. Implemented Functionality

2.1 Field Validation and Automation

Mandatory Fields

Mandatory fields are enforced on both the client and server side to prevent incomplete submissions from progressing through the workflow.

Auto-Populate Logic

Default values, such as the current user and location, are set automatically using Catalog Client Scripts and UI Policies, reducing manual data entry and the likelihood of user error.

Approval State Validation

The system ensures that a request cannot progress unless the required approval has been completed. This is enforced using flow conditions and approval state checks within the workflow, preventing requests from bypassing the approval stage.

2.2 Request Variables

The Network Request catalog item captures the following information from the requester. Ten variables are configured directly on the catalog item, in addition to a five-field Requester Information variable set:

Catalog Item Variables (10)

Question	Type	Notes
Address	Single Line Text	Order 1,000.
Connection Type	Multiple Choice	Order 700.
Device Details	Single Line Text	Order 1,100.
Device Type	Multiple Choice	Order 900. Choices: Laptop, Mobiles, Others.
Other Details	Single Line Text	Order 1,200. Shown only when Device Type = Others.
please provide device details	Single Line Text	Supplementary device description.
please_provide_address_here	Multi Line Text	Order 400. Mandatory delivery/installation address.
Relocated Address	Single Line Text	Order 800.
Requested For	Reference (User)	Order 1,300. Auto-populated with the current user.
type of device	Select Box	Order 300.

Requester Information Variable Set (5)

Name	Type	Question
email_id	Single Line Text	Email ID
opened_on_behalf_of	Reference (User)	Opened on behalf of
phone_number	Single Line Text	Phone Number
proof_of_document	Attachment	Proof of document
username	Single Line Text	UserName

2.3 Summary of Actions Taken

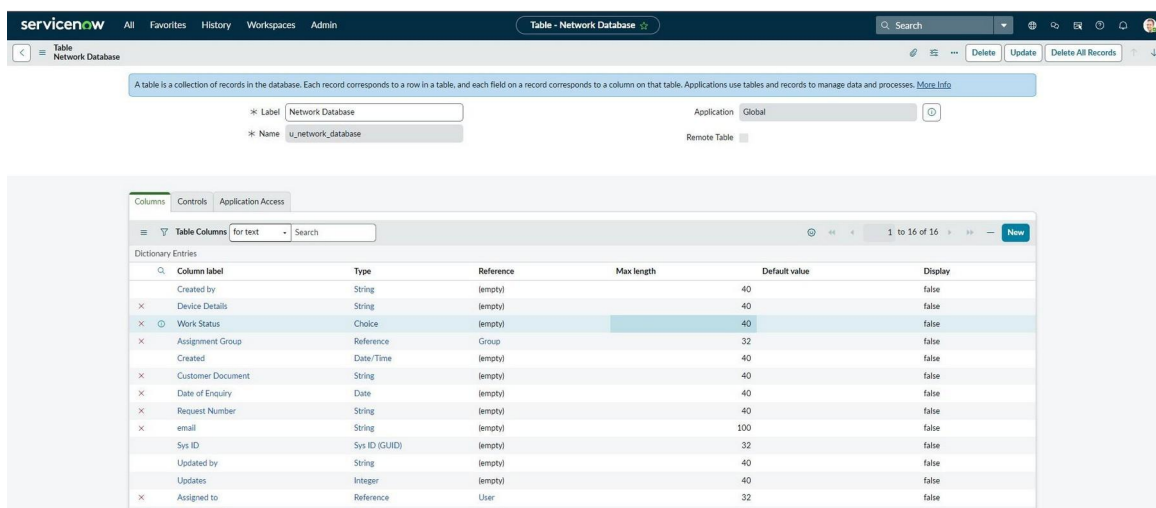
Area	Actions Taken
Data Handling	Catalog variables mapped to structured custom tables
Access Control	Role-based approvals and dynamic flow logic using Flow Designer
QA Testing	Full-cycle testing including request, approval, and notification logging
Data Integrity	Mandatory validations, auto-fills, and approval state enforcement

3. Data Model and Custom Tables

Catalog variables captured on the Network Request catalog item are mapped to structured custom tables, ensuring that request data is stored consistently and can be reliably queried and reported on. Custom tables were used to organise the request and task data: the Network Database table stores structured request information, while the Network Task table represents the work generated for the Network Team.

3.1 Custom Tables

- **u_network_database:** Stores structured network request information captured from the catalog item, including request details and associated metadata.
- **u_network_task:** Extends the base Task table and stores structured task-level information related to the fulfilment of network requests, including task status and assignment details.



Network Database (u_network_database) table and column configuration.

3.2 u_network_database Schema

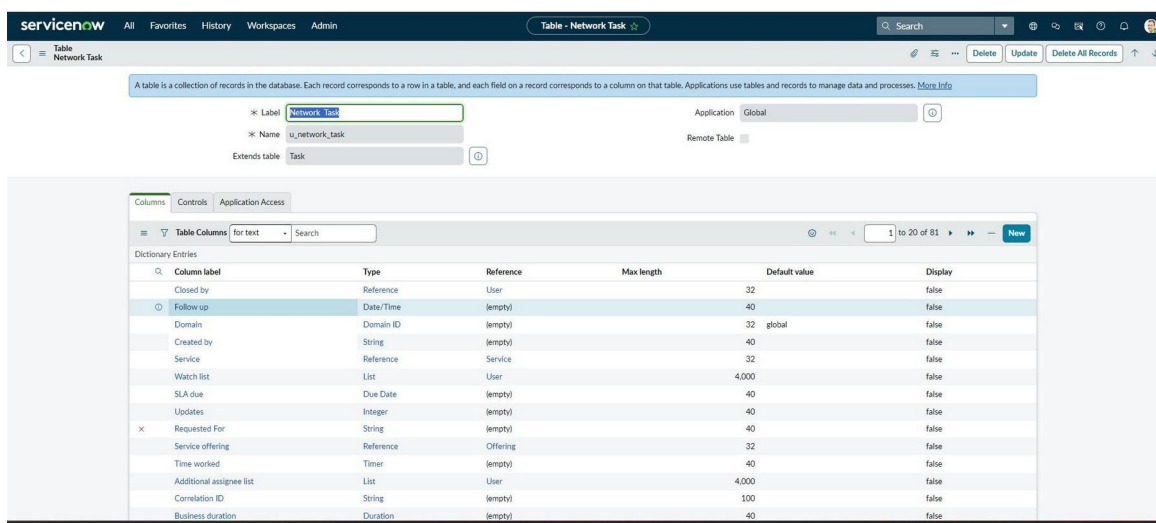
Key columns configured on the Network Database table include:

Column Label	Type	Notes
Request Number	String	Reference to the originating RITM.
email	String	Requester's email address captured from the catalog form.
Requested For	String	Populated from the catalog variable / current user.
Customer Address	String	Delivery or installation address.
Customer Document	String	Reference to any uploaded proof of document.
Device Details	String	Device information supplied by the requester.
Date of Enquiry	Date	Date the request was raised.
Work Status	Choice	Tracks the current status of the request (e.g.

Column Label	Type	Notes
		Requested, In Progress, Completed).
Assignment Group	Reference (Group)	Group responsible for fulfilment.
Assigned to	Reference (User)	Individual fulfiller assigned to the request.
Sys ID	Sys ID (GUID)	System-generated unique identifier.
Created / Created by	Date-Time / String	Audit fields populated automatically by the platform.
Updated / Updated by	Date-Time / String	Audit fields populated automatically by the platform.

3.3 u_network_task

The Network Task record is generated as part of the automated request lifecycle. It extends the base Task table (inheriting fields such as Requested For, Service Offering, Watch List, SLA due, and Business Duration) and carries the operational task information required by the Network Team, supporting subsequent status updates.



Column label	Type	Reference	Max length	Default value	Display
Closed by	Reference	User	32		false
Follow up	Date/Time	(empty)	40		false
Domain	Domain ID	(empty)	32	global	false
Created by	String	(empty)	40		false
Service	Reference	Service	32		false
Watch list	List	User	4,000		false
SLA due	Due Date	(empty)	40		false
Updates	Integer	(empty)	40		false
Requested For	String	(empty)	40		false
Service offering	Reference	Offering	32		false
Time worked	Timer	(empty)	40		false
Additional assignee list	List	User	4,000		false
Correlation ID	String	(empty)	100		false
Business duration	Duration	(empty)	40		false

Network Task (u_network_task) table, extending the base Task table.

3.4 Variable-to-Field Mapping

Each catalog variable defined on the Network Request catalog item is mapped to a corresponding field on the u_network_database and, where applicable, the u_network_task tables. This mapping logic ensures that data entered by the requester through the Service Portal is captured accurately and consistently within the data model.

3.5 Approval Request Relationship

An Approval Request relationship was configured so that approval information can be displayed against the Network Database record. The relationship connects the request record with approval records (sysapproval_approver) and provides an Approval Request related list directly on the Network Database form.

The screenshot shows the 'Relationship' configuration page for 'Approval Request'. The 'Name' field is set to 'Approval Request'. The 'Application' is 'Global'. The 'Applies to table' is 'Network Database [u_network_database]'. The 'Queries from table' is 'Approval [sysapproval_approver]'. A script is provided to refine the query in current that will populate the related list. The script is as follows:

```

1
2
3
4
(function refineQuery(current, parent) {
  current.addQuery('source_table', parent.getTableName());
  current.addQuery('document_id', parent.sys_id);
})(current, parent);

```

Below the script, there are buttons for 'Run Query Diagnostics', 'Update', and 'Delete'. Under 'Related Links', there is a link for 'Run Point Scan'. Under 'Updates (1)', there is a table showing the relationship configuration.

Created	Type	View	Target name	Name	Action	Update set	Remote update set	Updated by	Comments
2024-08-19 10:00:52	Relationship		Approval Request	sys_relationship_c8654c3d47e	INSERT_OR_UPDATE	Default	(empty)	admin	

At the bottom of the table, there is a 'Skip to row' button and a pagination indicator '1 to 1 of 1'.

Approval Request relationship configured against the Network Database table.

The screenshot displays a ServiceNow interface for a 'Network Database' record. The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', 'Workspaces', and 'Admin'. The record title is 'Network Database - Created 2026-08-22 01:52:01'. Below the title, there are several input fields for metadata: Request Number (RITM0010002), Date of Enquiry (2026-08-20), Assignment Group (Network), Customer Address (patna), Customer Document, Work Status (Completed), Assigned to (Adela Cervantesz), Device Details (laptop), and Requested For. An 'email' field is also present. Below these fields, there is an 'Approvals' section with a table. The table has columns for 'State', 'Approver', 'Comments', 'Approval for', and 'Created'. A single row is visible with the state 'Approved', approver 'System Administrator', and creation date '2026-08-22 01:52:01'. The table is paginated to show '1 to 1 of 1' rows.

Approval Request related list displayed directly on a Network Database record.

3.6 Data Integrity Measures

- Mandatory validation of required fields before submission.
- Auto-filled values for fields such as Requested For and requester.
- Approval state enforcement to prevent premature progression.
- Structured table mapping between catalog variables and custom tables.

4. Approval Use Cases

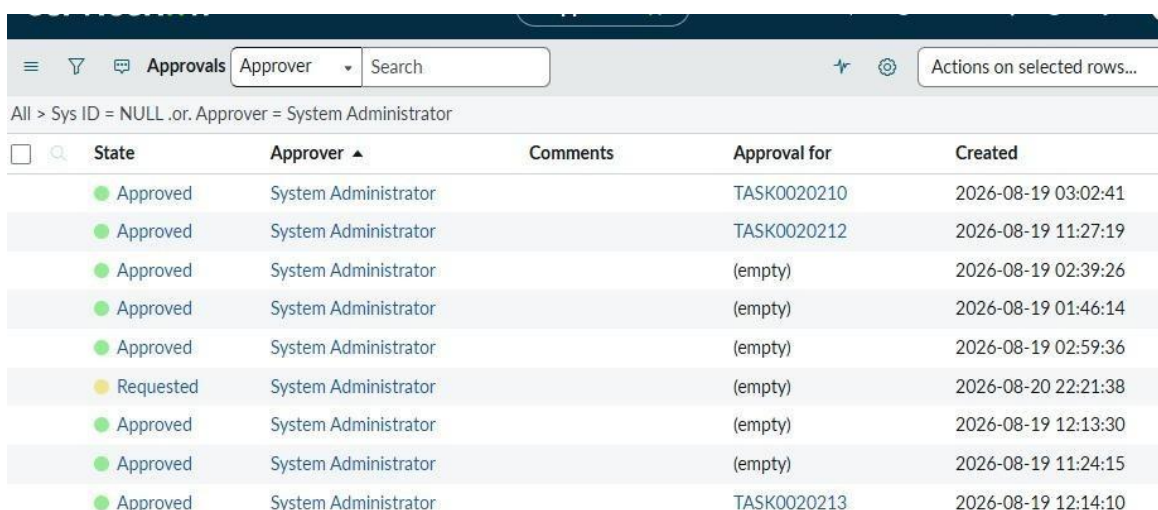
Approval routing in the Network Request Management system is dynamic and depends on the characteristics of the request, such as its sensitivity and the department it originates from.

#	Approval Scenario	Description
1	Manager Approval	Applied to standard network requests requiring routine managerial sign-off.
2	Network Security Approval	Applied to high-sensitivity requests that require review by the Network Security team.
3	Group Approval	Applied to department-specific tasks that require sign-off from a designated approval group.

The specific approval path is determined by conditions evaluated within Flow Designer, ensuring that each request is routed to the appropriate approver without manual intervention.

4.1 Approval Output

The approval output confirms that an approval record is generated against the request and can reach an Approved state with an approver associated. Approval records also link back to the Network Task once fulfilment begins, so that the full approval trail — from initial request through to task-level approval — remains visible and auditable.



The screenshot shows the 'Approvals' table in ServiceNow. The table has columns for State, Approver, Comments, Approval for, and Created. The filter is set to 'All > Sys ID = NULL .or. Approver = System Administrator'. The table contains 10 rows of data, with 9 'Approved' records and 1 'Requested' record. All approvers are 'System Administrator'.

State	Approver	Comments	Approval for	Created
Approved	System Administrator		TASK0020210	2026-08-19 03:02:41
Approved	System Administrator		TASK0020212	2026-08-19 11:27:19
Approved	System Administrator		(empty)	2026-08-19 02:39:26
Approved	System Administrator		(empty)	2026-08-19 01:46:14
Approved	System Administrator		(empty)	2026-08-19 02:59:36
Requested	System Administrator		(empty)	2026-08-20 22:21:38
Approved	System Administrator		(empty)	2026-08-19 12:13:30
Approved	System Administrator		(empty)	2026-08-19 11:24:15
Approved	System Administrator		TASK0020213	2026-08-19 12:14:10

Approval records showing approval state and approver.

5. Automation and Flow Designer

The project uses Flow Designer for end-to-end automation of the network request lifecycle. A flow named “Network Request” orchestrates the process from submission through to fulfilment and closure, triggered by the Service Catalog.

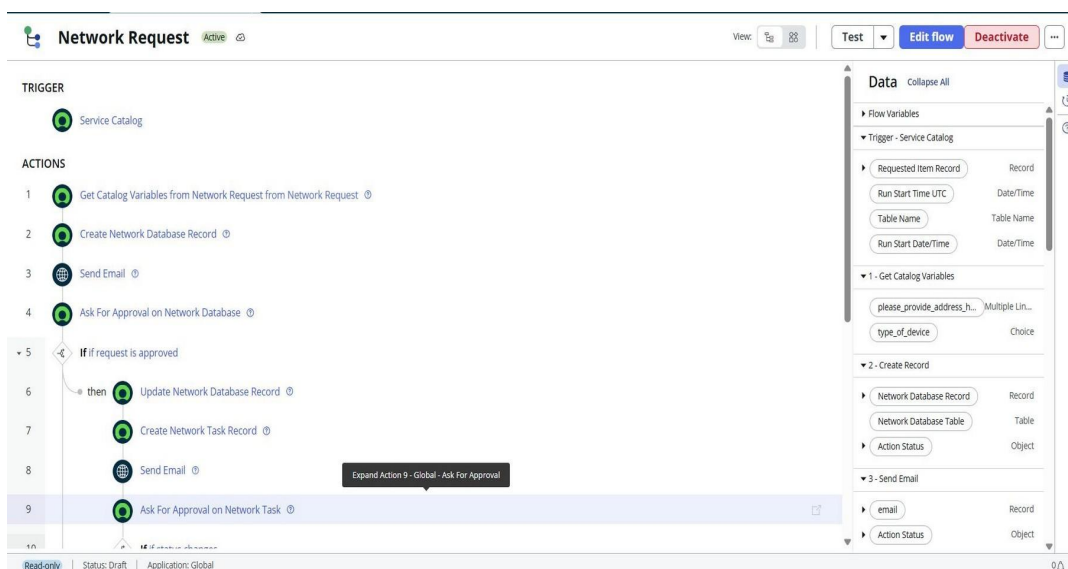
5.1 Flow Trigger and Actions

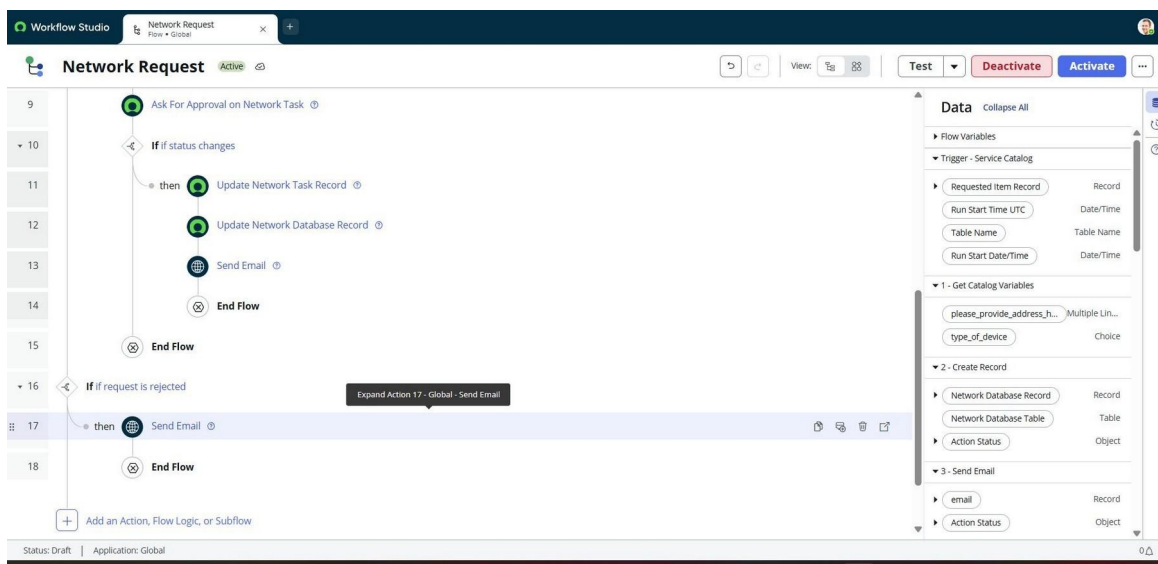
The Network Request flow is triggered on Service Catalog submission and executes the following sequence of actions:

1. Get Catalog Variables from Network Request – retrieves the submitted variable values (e.g.

Please Provide Address Here, Type of Device).

2. Create Network Database Record – creates the corresponding record on u_network_database using the retrieved variables.
3. Send Email – notifies the requester that the request has been created.
4. Ask For Approval on Network Database – raises the approval request against the newly created record.
5. If request is approved – branches the flow based on the approval outcome:
 - Update Network Database Record
 - Create Network Task Record
 - Send Email
 - Ask For Approval on Network Task
6. If status changes – when the Network Task status changes, the flow:
 - Updates the Network Task Record
 - Updates the Network Database Record
 - Sends Email
7. If request is rejected – the flow sends a rejection Email and ends.





Flow Designer – approval and rejection branches, and status-change handling.

5.2 Flow Capabilities Used

- Dynamic approvals based on request characteristics.
- Dynamic task creation for fulfilment.
- Approval condition logic implemented within Flow steps (If request is approved / If request is rejected / If status changes).
- Notification triggers at key stages of the workflow.
- Request status progression as the flow advances.
- Flow execution logging for traceability and troubleshooting.

5.3 Implementation Approach

A key implementation detail of this project is that the solution uses Flow Designer and does not depend on unnecessary custom Script Includes. This design choice was intentional:

- Flow Designer is used for dynamic approvals and task creation, keeping automation logic centralised and visible.
- Catalog Client Scripts and UI Policies are used only for essential, minimal form-level behaviour.
- Business Rules are limited to supporting configuration rather than driving the core process,

since the core process is orchestrated by Flow Designer.

This approach ensures maintainability and ease of handoff to administrators or future developers, since the majority of the process logic is visible and configurable directly within Flow Designer rather than embedded in custom server-side scripts.

6. Form Behaviour and UI

Phase 3 of the project focused on creating a user-friendly Service Portal experience for Network Request submission, defining the navigation path, improving usability, and applying business/UI logic. The Network Request Catalog Item is configured with the following components:

Component	Count	Purpose
Variables	10	Collect the required network request information from the user.
Variable Set	1	Groups and manages related variables (Requester Information: Email ID, Opened on Behalf Of, Phone Number, Proof of Document, UserName).
Catalog UI Policy	1	Controls the visibility or behaviour of fields based on user selections.
Catalog Client Script	1	Performs client-side actions when the catalog form is loaded or interacted with.
Catalog Availability	—	Controls where and for whom the Network Request catalog item is available.

6.1 Catalog Item Configuration Evidence

The Network Request catalog item and its supporting configuration were built through the following activities.

Activity 1: Creation of the Service Catalog Item

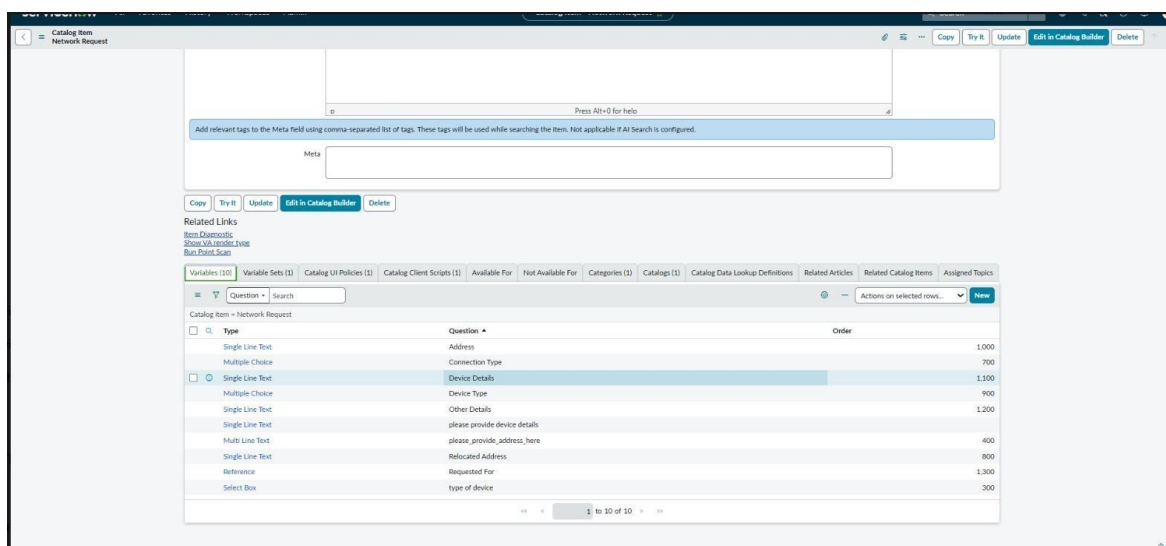
Navigation: Service Catalog → Catalog Definitions → Maintain Items → New. The Network Request catalog item was created under the Network category, with the short description “Network Request Management.”

The screenshot shows the ServiceNow 'Catalog Item - Network Request' configuration page. The top navigation bar includes 'All', 'Favorites', 'History', 'Workspaces', and 'Admin'. The breadcrumb trail is 'Catalog Item - Network Request'. The page title is 'Catalog Item - Network Request'. A blue banner at the top states: 'Build and modify items faster with the improved Catalog Builder. Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies. Enter a Name and Short description to display for the item. Enter a Price, approvals, variables, and other information as needed.' Below the banner, the configuration fields are: Name: 'Network Request', Catalog: 'Service Catalog', Category: 'Network', State: 'None', Checked out: 'None', Owner: 'System Administrator', Application: 'Global', Active: 'Active', and Fulfillment automation level: 'Unspecified'. The 'Item Details' tab is selected, showing the 'Short description' field with the value 'Network Request Management'. The 'Description' field is empty. The 'Process Engine', 'Picture', 'Pricing', and 'Portal Settings' tabs are also visible.

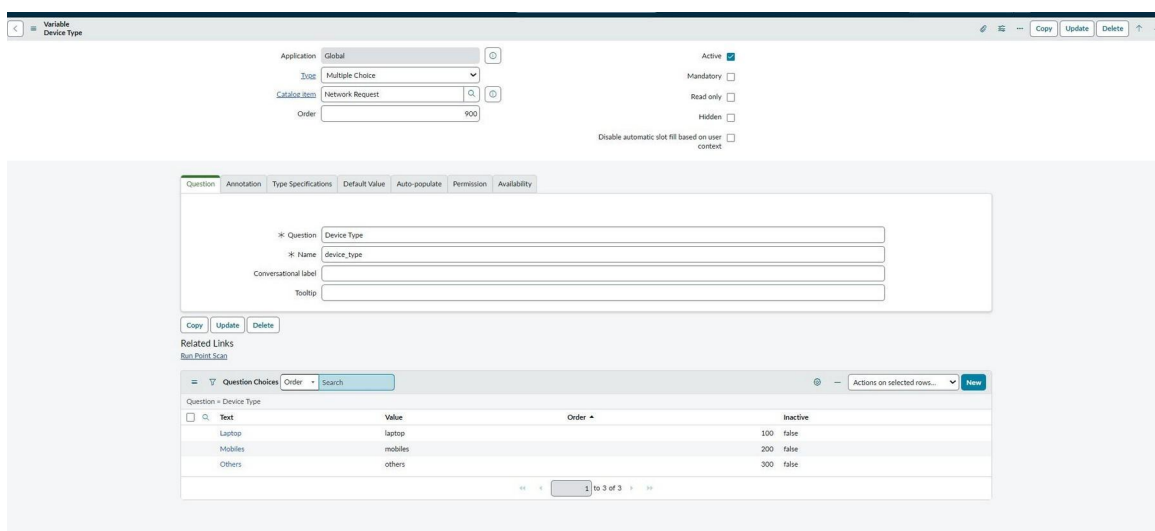
Network Request catalog item created under the Service Catalog.

Activity 2 & 3: Variables Configuration and Creation

Ten variables were created and ordered on the catalog item, with mandatory fields and choice lists (such as Laptop, Mobiles, and Others for Device Type) configured as required.



Variables configured on the Network Request catalog item (10 variables).



Device Type variable with configured choices: Laptop, Mobiles, and Others.

Activity 4: Variable Set Configuration

Navigation: Service Catalog → Variable Sets. A Requester Information variable set was created and associated with the Network Request catalog item, grouping five related variables into a single-row, two-column layout.

Name	Type	Question	Order
email_id	Single Line Text	Email ID	
opened_on_behalf_of	Reference	Opened on behalf of	
phone_number	Single Line Text	Phone Number	
proof_of_document	Attachment	Proof of document	
username	Single Line Text	Username	

Requester Information variable set, grouping five related variables.

6.2 UI Policy Configuration

A Catalog UI Policy named “Types of devices is others” is configured for the Network Request catalog item. This policy controls the form behaviour based on the selected device type: it is configured to run On Load and automatically reverses its behaviour when the condition becomes false, so that additional fields (such as Other Device Details) only appear when the requester selects “Others” as the device type.

Navigation: Service Catalog → Catalog Definitions → Maintain Items → Network Request → Catalog Client Scripts

Short description	Variable set	Conditions	Reverse if false	On load	Inherit	Updated	Order
Types of devices is others	(empty)		true	true	false	2024-08-21 01:29:12	100

Catalog UI Policy “Types of devices is others” and related client-side configuration.

6.3 Navigation Flow

The user navigates from the ServiceNow Service Portal to the Network Request catalog item, enters the required information, and submits the request. A request number is generated and the workflow continues with automated processing and notifications.

- Log in to the ServiceNow portal.
- Open the Service Portal and search for Network Requests.
- Open the Network Request catalog item.
- Fill in the required details.
- Submit the request.
- Use the generated request number to track the request and receive notifications.

6.4 Usability

The interface groups related information into a single request form and provides visible mandatory-field indicators, submission controls, Requested For information, and attachment support. Required fields such as Address and Device Details are clearly flagged in the sidebar of the Service Portal form.

The screenshot displays the ServiceNow 'Network Request' form. The header includes the ServiceNow logo and navigation links: Knowledge, Catalog, Requests, System Status, Cart, Tours, and a user profile for 'System Administrator'. The breadcrumb trail shows 'Home > Service Catalog > Network > Network Request'. A search bar is present. The main form area is titled 'Network Request' and 'Network Request Management'. It contains several input fields: 'Email ID' (filled with 'abraham.lincoln@example.com'), 'Phone Number' (filled with '(555) 555-0004'), 'type of device' (set to 'Laptop'), and 'Opened on behalf of' (set to 'Abraham Lincoln'). There is an 'Upload' button for 'Proof of document' and a 'UserName' field (filled with 'Abraham Lincoln'). The right sidebar features a 'Quantity' dropdown (set to 1), 'Delivery Time: 0 Days', 'Add to Cart', 'Save as Draft', and 'Order Now' buttons. Below these are 'Required information' links for 'Address' and 'Device Details'. The bottom of the form has a 'please_provide_address_here' field.

Network Request form completed with sample requester information.

Form UI behaviour was verified through Service Portal form tests and preview mode, ensuring that field visibility, mandatory rules, and auto-population worked correctly before the catalog item was made available to end users.

7. Troubleshooting

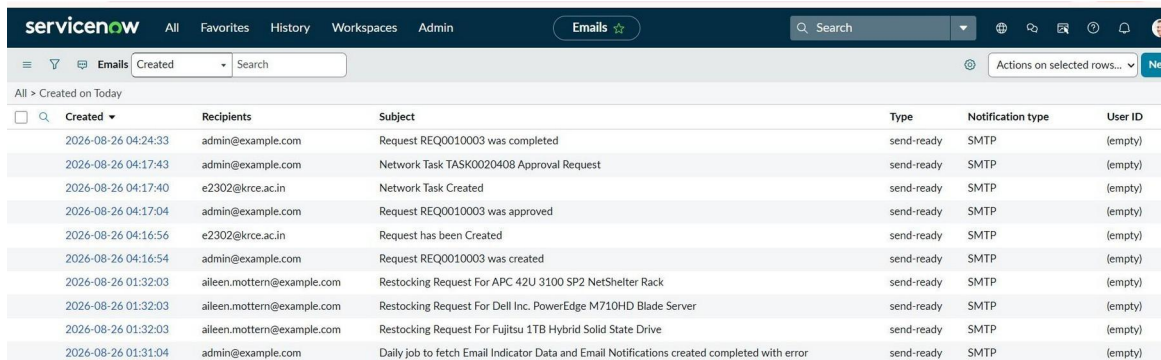
Throughout development and testing, several troubleshooting techniques were used to diagnose and resolve issues in the request flow, form behaviour, and notifications.

Flow Execution Logs

Flow Designer's Execution Details were used to trace flow steps, identify failures, and verify condition paths during approvals and notifications. This allowed each stage of the flow to be inspected individually to confirm that the correct branch was taken and that data was passed correctly between steps (see the flow views in Section 5.1).

Email Logs

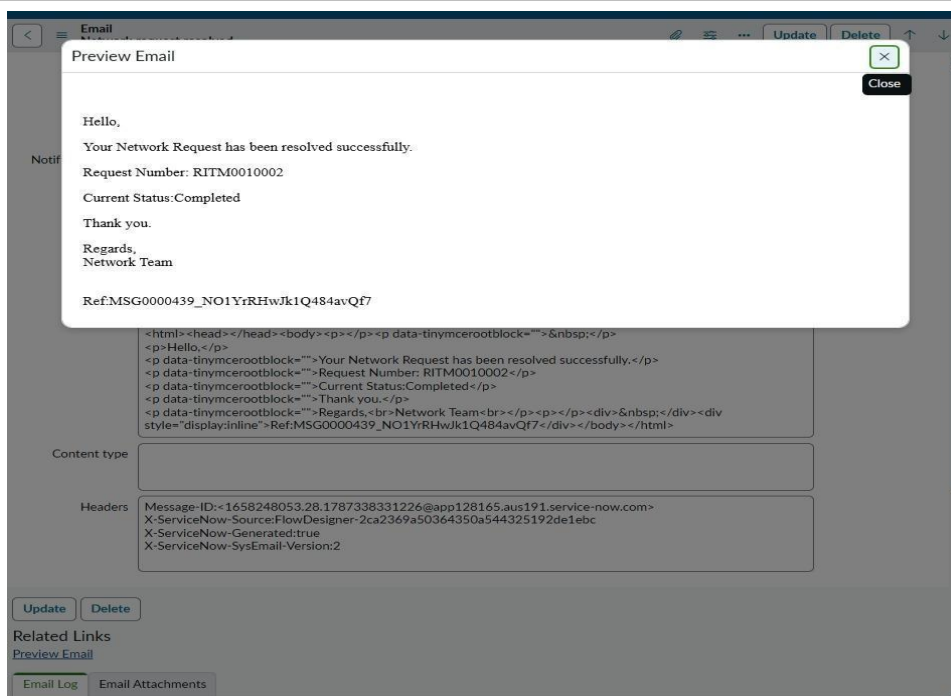
Email logs were checked under System Logs → Email → Sent/Received to validate notification triggers and message delivery status. This confirmed that notifications were generated at the correct points in the workflow and delivered to the intended recipients.



The screenshot shows the ServiceNow interface with the 'Emails' tab selected. The table displays the following data:

Created	Recipients	Subject	Type	Notification type	User ID
2026-08-26 04:24:33	admin@example.com	Request REQ0010003 was completed	send-ready	SMTP	(empty)
2026-08-26 04:17:43	admin@example.com	Network Task TASK0020408 Approval Request	send-ready	SMTP	(empty)
2026-08-26 04:17:40	e2302@krce.ac.in	Network Task Created	send-ready	SMTP	(empty)
2026-08-26 04:17:04	admin@example.com	Request REQ0010003 was approved	send-ready	SMTP	(empty)
2026-08-26 04:16:56	e2302@krce.ac.in	Request has been Created	send-ready	SMTP	(empty)
2026-08-26 04:16:54	admin@example.com	Request REQ0010003 was created	send-ready	SMTP	(empty)
2026-08-26 01:32:03	alileen.mottern@example.com	Restocking Request For APC 42U 3100 SP2 NetShelter Rack	send-ready	SMTP	(empty)
2026-08-26 01:32:03	alileen.mottern@example.com	Restocking Request For Dell Inc. PowerEdge M710HD Blade Server	send-ready	SMTP	(empty)
2026-08-26 01:32:03	alileen.mottern@example.com	Restocking Request For Fujitsu 1TB Hybrid Solid State Drive	send-ready	SMTP	(empty)
2026-08-26 01:31:04	admin@example.com	Daily job to fetch Email Indicator Data and Email Notifications created completed with error	send-ready	SMTP	(empty)

Email log entries confirming notification triggers across the request lifecycle.



Preview of an automatically generated email confirming request resolution.

Variable Population Issues

Misaligned or missing data bindings between catalog variables and table fields were resolved using:

- Variable set debugging to identify incorrect or missing mappings.
- Catalog Client Scripts for dynamic population of dependent fields.

Form Behaviour Testing

UI Policies and visibility logic were confirmed via Service Portal form tests and preview mode, ensuring that the form behaved as intended across different request scenarios before deployment to end users.

General Troubleshooting Procedure

1. Review the complete Network Request workflow.
2. Check for errors in the catalog item, variables, approvals, tasks, and notifications.
3. Verify that the configured flow executes correctly.
4. Resolve any configuration or validation issues found.
5. Perform a final end-to-end verification.

8. Testing and Security

Full-cycle testing was carried out to validate the Network Request Management system end to end, covering functional, data-integrity, and security aspects of the solution.

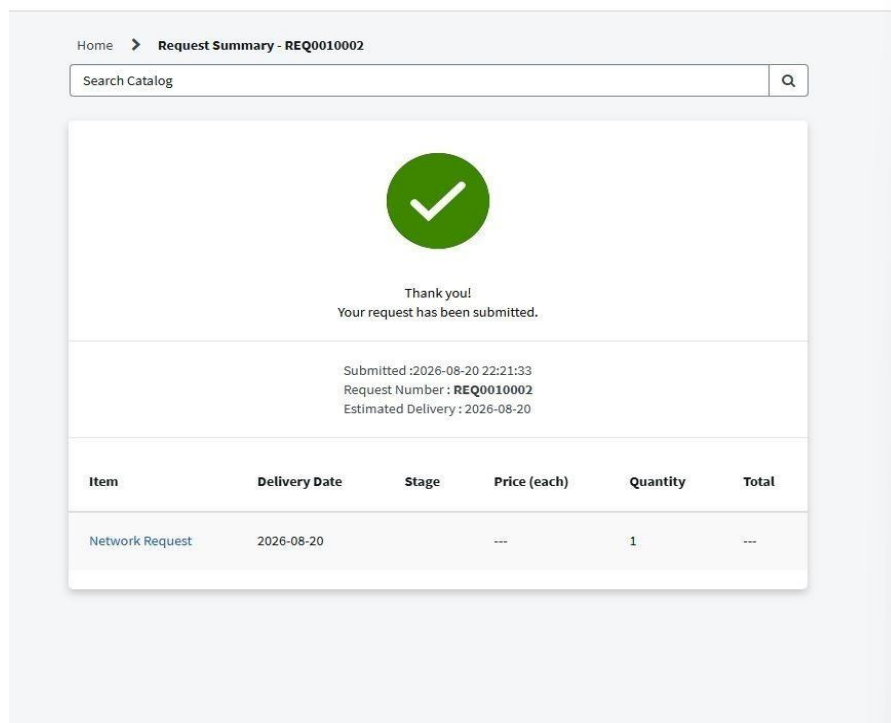
8.1 Areas Covered in Testing

- Request submission
- Mandatory fields
- Dynamic fields
- Auto-population
- Approval routing
- Approval state validation
- Task creation
- Email notifications
- Request status
- Flow execution
- Data mapping
- Role-based access
- ACL behaviour
- UI Policies
- Catalog Client Scripts

8.2 QA Test Procedure

End-to-end testing covered request submission, record creation, approval processing, task creation, status updates, and email notification, using the following procedure:

1. Open the Network Request catalog item.
2. Enter valid request details.
3. Submit the request using Order Now.
4. Verify that the Request/RITM is generated.
5. Verify the approval request is triggered.
6. Approve the request.
7. Verify that the Network Task is created.
8. Complete the task and verify the request status.
9. Check that the configured notifications are generated.



Service Portal confirmation screen following successful request submission, with generated request number.

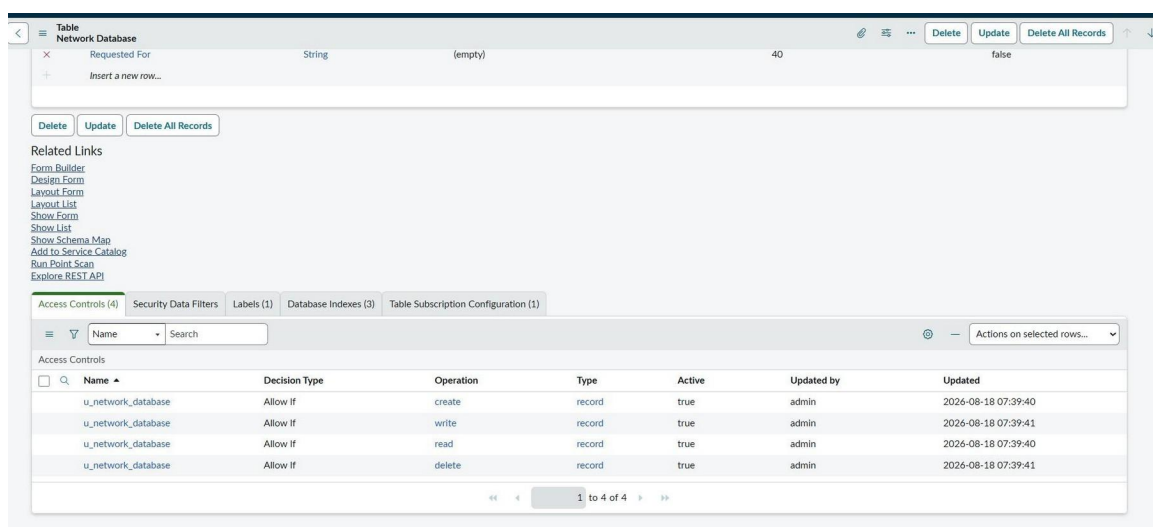
Testing was carried out using realistic use cases and sample data representative of typical network request scenarios, ensuring the solution behaves reliably under conditions similar to real-world usage.

8.3 Security and Access Control

Security in the Network Request Management system is handled through role-based access control and ACL configuration. Access to the `u_network_database` table is governed by four active Access Control rules covering the create, read, write, and delete operations, each set to an “Allow If” decision type:

Table	Operation	Decision Type	Type
<code>u_network_database</code>	create	Allow If	record
<code>u_network_database</code>	write	Allow If	record
<code>u_network_database</code>	read	Allow If	record
<code>u_network_database</code>	delete	Allow If	record

Access to catalog items, request records, approval actions, and fulfilment tasks is restricted according to the user's assigned role, ensuring that only authorised users can view or act on specific data.



Access Control (ACL) rules configured for the Network Database table.

9. Project Execution Roadmap

The project was executed across five phases, moving from initial planning through to final deployment and presentation:

Requirement Analysis → Backend Configuration → UI/UX Customization → Automation & Security → Testing → Documentation & Deployment

Phase	Description
Phase 1	Requirement Analysis & Planning
Phase 2	Backend Development & Configurations
Phase 3	UI/UX Development & Customization
Phase 4	Data Migration, Testing & Security
Phase 5	Deployment, Documentation & Final Presentation

9.1 Execution Sequence

1. Catalog Creation
2. Form Setup
3. Approval Integration
4. Testing & Security
5. Deployment

10. Phase 1 – Requirement Analysis & Planning

Phase 1 established the foundation for the project by identifying the problem, defining the required ServiceNow components, identifying stakeholders, and preparing the roadmap.

10.1 Understanding the Requirement

- Identify the problem: network service requests require a structured and automated process.
- Decide to automate the request lifecycle using ServiceNow.
- Define the main objective of the system.
- Identify the information that users need to provide while submitting a network request.
- Define the overall process: Request → Validation → Approval → Network Task → Fulfilment → Status Update → Notification.

10.2 ServiceNow Components Identified

- Service Catalog
- Network Request Catalog Item
- Catalog Variables
- Network Database Table
- Flow Designer
- Approval Request
- Network Task
- Catalog Client Script
- Business Rules
- Notifications
- Access Controls
- Update Sets

10.3 Phase 2 – Backend Development Summary

Phase 2 established the backend structure for Network Requests, Network Database records, Network Tasks, and approval handling. This phase produced the custom tables and the Approval Request relationship documented in Section 3 of this document.

10.4 Phase 3 – UI/UX Development Summary

Phase 3 focused on creating a user-friendly Service Portal experience for Network Request submission, defining the navigation path, improving usability, and applying business/UI logic. This phase produced the catalog item, variables, variable set, and UI Policy documented in Section 6 of this document.

11. Phase 4 – Data Migration, Testing & Security

Phase 4 validated the automated workflow, protected the custom data using Access Controls, tested the request lifecycle, and verified data integrity.

11.1 Automation

The Network Request process was automated using ServiceNow Flow Designer. The flow obtains catalog variables, creates the Network Database record, sends the approval request, handles approved and rejected branches, creates and updates Network Tasks, updates the Network Database status, and sends email notifications.

- Open ServiceNow → All → Flow Designer.
- Open the configured Network Request flow.
- Verify the request trigger and workflow actions.
- Configure the required approval and fulfilment actions.
- Verify that the Network Task is created after approval.
- Save and activate the flow.

11.2 Access Control

Access control was implemented through active ACL configurations. Read, create, write, and delete operations are controlled so that access to the custom data is restricted according to the configured rules.

- Navigate to All → System Security → Access Control (ACL).
- Verify access permissions for the Network Request data.
- Ensure only authorised users can create or update requests.
- Verify that approvers and fulfilment users have the required access.
- Save the ACL configuration.

11.3 QA Testing

End-to-end testing covered request submission, record creation, approval processing, task creation, status updates, and email notification. The request submission output confirms successful request creation and generation of a request number (see Section 8.2 for the full test procedure).

11.4 Data Integrity

Data integrity was supported through mandatory fields, automatic requester population, and approval-state checks. The Service Portal form demonstrates required information and automatic Requested For population.

- Open the generated Network Request record.
- Verify that all submitted values are stored correctly.
- Check the relationship between the request, approval, and task records.
- Verify that no required information is missing or incorrectly mapped.
- Confirm that the complete request lifecycle is properly recorded.

Troubleshooting during this phase was performed primarily using Flow Designer Execution Details and ServiceNow email logs, which provided clear visibility into flow step outcomes and notification delivery status.

12. Phase 5 – Deployment, Documentation & Final Presentation

Phase 5 is the final documentation and presentation stage of the project. It covers deployment of the final solution, preparation of technical and user documentation, and presentation of the working system to stakeholders, structured across eleven activities.

12.1 Troubleshooting

1. Review the complete Network Request workflow.
2. Check for errors in catalog item, variables, approvals, tasks, and notifications.
3. Verify that the configured flow executes correctly.
4. Resolve any configuration or validation issues.
5. Perform a final end-to-end verification.

12.2 Adherence to Timelines

1. Review the planned project phases and activities.
2. Verify that all configured components are completed.
3. Ensure testing and deployment activities are completed within the planned timeline.
4. Track pending issues and complete the required corrections.

12.3 Innovation

1. Highlight the automation introduced in the Network Request process.
2. Show how manual request handling is reduced.
3. Demonstrate automated approval and task creation.
4. Explain how ServiceNow improves request tracking and visibility.

12.4 Document Functional Overview

1. Document the purpose of the Automated Network Request Management system.
2. Explain the complete request lifecycle.
3. Describe the Service Catalog, variables, approvals, tasks, and notifications.
4. Document the expected user and fulfilment-team workflow.

12.5 Document Technical Blueprint

1. Document the ServiceNow components used.
2. Explain the Network Request catalog item.
3. Document the custom tables such as Network Database and Network Task.
4. Document variables, variable sets, approvals, Flow Designer, and related configurations.
5. Describe the relationship between request, approval, and fulfilment records.

12.6 Document Setup Manual

A step-by-step manual is provided for recreating the solution in a new Personal Developer Instance (PDI):

1. Document the navigation required to access each configuration.
2. Provide the steps for creating the catalog item and variables.
3. Document table, approval, automation, and security configuration.
4. Provide the testing procedure.
5. Include the required screenshots as configuration evidence.

12.7 Project Demo Video Planning

1. Prepare the demonstration sequence.
2. Start with the Network Request catalog item.
3. Show request submission and generated records.
4. Demonstrate approval and task creation.
5. Show the final request status and tracking.

12.8 Visual Demonstration

1. Demonstrate the configured ServiceNow interface.
2. Show the Network Request form.
3. Show the approval and fulfilment process.
4. Demonstrate the final output using actual ServiceNow records.

12.9 Explanation Clarity

1. Explain each configuration in a clear sequence.
2. Describe the purpose of every major component.
3. Explain how the request moves from submission to completion.
4. Keep the final presentation aligned with the implemented system.

12.10 Realism & Quality

1. Demonstrate the actual configured ServiceNow system.
2. Verify that the screenshots and records represent the implemented configuration.
3. Ensure the workflow produces the expected results.
4. Present the final system in a clear and professional manner.

12.11 Scalability & Future Plan

1. Explain how the Network Request solution can be extended to additional network services.
2. Add more request types and approval levels when required.
3. Extend automation for additional fulfilment activities.
4. Improve reporting and monitoring in future versions.

13. End-to-End Functional Output

1. User opens the Network Request through the Service Portal.
2. Requester and network/device details are entered in the configured request form.
3. The user submits the request using Order Now, and ServiceNow generates a Request (REQ) and Requested Item (RITM).
4. The submitted request details are recorded in the configured Network Database record.
5. An approval request is automatically generated for the configured approver.
6. The approver reviews the request and approves or rejects it.
7. When the request is approved, a Network Task is automatically created for the fulfilment team.
8. The Network Task and Network Database status are updated as the request progresses through the configured workflow.
9. After successful completion, the request is marked as completed/resolved, and a completion notification email is generated for the configured recipient.

13.1 Final Workflow

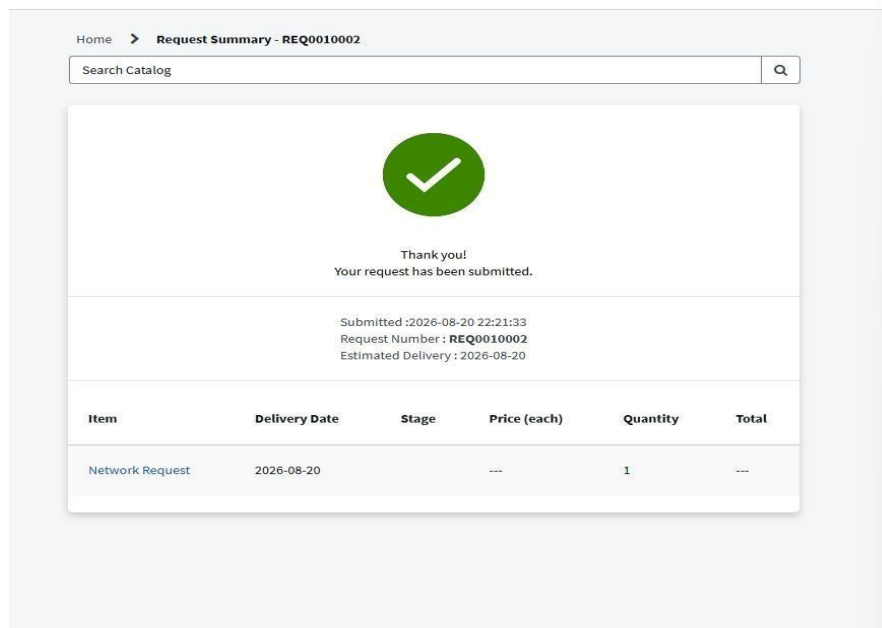
Service Portal → Network Request → REQ/RITM → Network Database → Approval → Network Task → Task Completion → Status Update → Email Notification

13.2 Functional Output Evidence

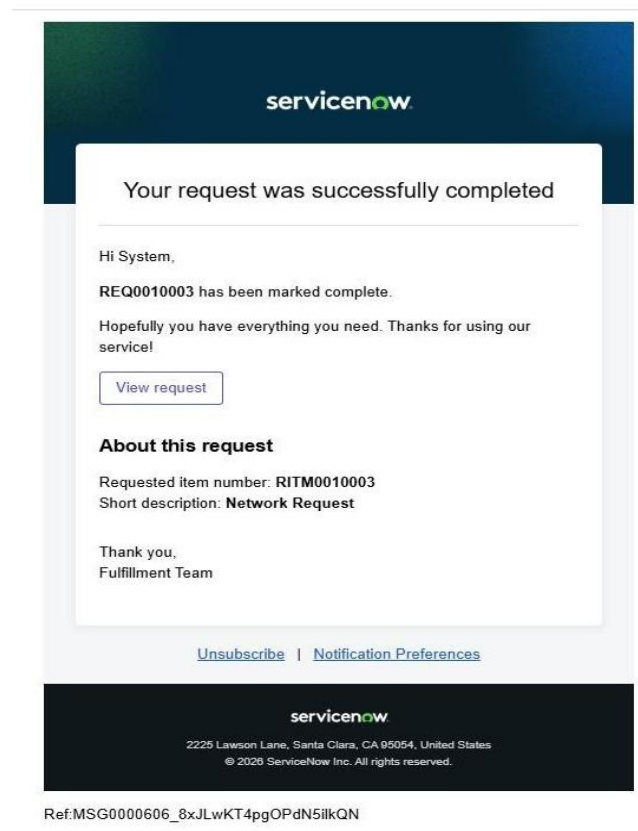
Completed request form showing required fields and requester information.

	State	Approver	Comments	Approval for	Created
	Approved	System Administrator		TASK0020210	2026-08-19 03:02:41
	Approved	System Administrator		TASK0020212	2026-08-19 11:27:19
	Approved	System Administrator		(empty)	2026-08-19 02:39:26
	Approved	System Administrator		(empty)	2026-08-19 01:46:14
	Approved	System Administrator		(empty)	2026-08-19 02:59:36
	Requested	System Administrator		(empty)	2026-08-20 22:21:38
	Approved	System Administrator		(empty)	2026-08-19 12:13:30
	Approved	System Administrator		(empty)	2026-08-19 11:24:15
	Approved	System Administrator		TASK0020213	2026-08-19 12:14:10

Approval records confirming the request's approval trail.



Service Portal confirmation screen following successful request submission.



Branded completion email sent to the requester on request closure.

14. Deployment

Prior to making the solution available to end users, the following verification and testing activities were completed:

- Final configuration verification
- Flow verification
- Catalog item verification
- Variable verification
- ACL and role verification
- Approval testing
- Notification testing
- End-to-end testing
- Final documentation
- Demo preparation

All configuration changes made throughout the project were tracked using ServiceNow Update Sets, allowing the complete set of customisations — the catalog item, variables, variable set, UI Policy, custom tables, Flow Designer flow, and Access Controls — to be captured and moved as a single, reviewable unit.

Note: This project was implemented and demonstrated within a ServiceNow Personal Developer Instance (PDI). No production deployment has taken place. All configuration, testing, and demonstration activities described in this document refer to the PDI/demo environment, and any future production rollout would require a separate, formal deployment and change-management process.

15. User Documentation

15.1 Requester / End User

1. Log in to the ServiceNow portal.
2. Open the Service Portal and navigate to Network Services.
3. Open the Network Request catalog item and complete the Requester Information and request details.
4. Submit the request using Order Now.
5. Track the request status using the generated request number.

15.2 Manager / Approver

1. Receive the approval notification.
2. Review the request and check the request details.
3. Approve or reject the request according to role.

15.3 Network Fulfilment Team

1. Receive the generated Network Task.
2. Process the network-related activity described in the task.
3. Update the task/request status as work progresses.
4. Complete fulfilment and close the task.

15.4 ServiceNow Administrator

1. Configure the catalog item, variables, and variable sets.
2. Configure Flow Designer, Business Rules, Access Controls, and Notifications.
3. Monitor flow execution and email logs for issues.
4. Maintain the solution and apply Update Sets for future changes.

16. System Benefits

- Reduced manual effort
- Faster request processing
- Consistent approval handling
- Better transparency
- Improved data integrity
- Automated notifications
- Better auditability
- Role-based security
- Maintainability
- Scalability

17. Conclusion

The Automated Network Request Management in ServiceNow project successfully implements a structured and automated process for managing network-related service requests. The system provides a complete lifecycle: Request Submission → Request Creation → Approval → Network Task → Fulfilment → Status Update

* Notification.

The project uses the Service Catalog, Catalog Variables, Variable Sets, custom tables (u_network_database and u_network_task), Approvals, Flow Designer, Network Tasks, Business Rules, UI logic, and Notifications to reduce manual effort and improve request tracking. By centralising process logic in Flow Designer rather than relying on custom Script Includes, the solution remains transparent and straightforward for administrators and future developers to maintain.

Approvals are controlled dynamically, with routing determined by request characteristics, ensuring that Manager, Network Security, or Group approval is applied where appropriate. Mandatory field

validation, auto-population, and approval-state enforcement improve the quality and consistency of data captured at intake, while structured mapping to the `u_network_database` and `u_network_task` custom tables ensures that this data is stored in a reliable, queryable form.

The final implementation was validated through the end-to-end functional output described in Section 13: the request was successfully submitted, an approval was generated and processed, the fulfilment process was completed, and the requester received a completion notification.

17.1 Key Outcomes

- Centralized network request submission
- Structured request information through catalog variables
- Automated approval process
- Automated Network Task creation
- Improved request tracking and visibility
- Reduced manual intervention
- Status updates throughout the request lifecycle
- Completion notification to the requester

Looking ahead, the architecture can be extended through integrations with network tools such as Cisco DNA Centre and Ansible for auto-provisioning, orchestration via ServiceNow Integration Hub, reporting dashboards for operational metrics, and expansion to cover additional IT services such as device provisioning and access requests. These enhancements would build on the existing Flow Designer-based foundation without requiring significant rework of the core solution.

18. Final Project Summary

Project

Automated Network Request Management in ServiceNow

Core Problem

Manual and inconsistent network request handling can lead to delays, incomplete information, approval gaps, and poor visibility.

Solution

A ServiceNow-based automated request management system using the Service Catalog, dynamic forms, Flow Designer, role-based approvals, ACLs, email notifications, custom tables, validation, testing, and status tracking.

Core Workflow

Request → Validation → Routing → Approval → Task Creation → Notification → Fulfilment → Tracking

Key Technologies

- ServiceNow Service Catalog
- Flow Designer
- Catalog Client Scripts
- UI Policies
- ACLs
- Roles
- Custom Tables (u_network_database, u_network_task)
- Email Notifications
- Service Portal
- Update Sets

Future Scope

- Network tool integrations (e.g. Cisco DNA Centre, Ansible)
- Auto-provisioning
- Integration Hub orchestration
- Reporting dashboards
- Expansion to other IT service